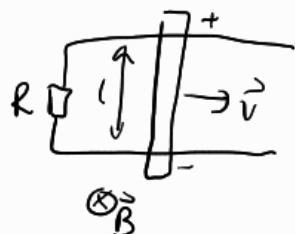


Indukcija

I v zanki $\vec{B} \rightarrow \vec{M}$ $\in$ l. motor $W_{el} \rightarrow W_{meh}$

$\vec{M}$ na zanko $\vec{B} \rightarrow I$? $\checkmark$ $\in$ l. generator $W_{meh} \rightarrow W_{el}$

Generator enasm. nap.



$$U_i = U_H = Blv$$

za $U = 1V$: $l = 1m$, $v = 1m/s$

Magnetni pretok

$$\Phi_m = \int \vec{B} \cdot d\vec{S}$$

za zg. primer

$$\text{Faraday: } U_i = \frac{d\Phi_m}{dt} = \frac{d(BS)}{dt} = B \frac{dx \cdot l}{dt} = Blv \checkmark$$

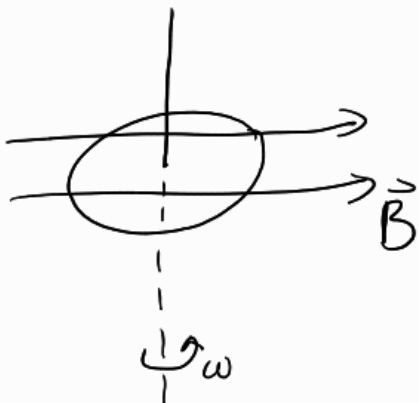
↓
zares je to
mešani produkt
 $(\vec{B}, \vec{l}, \vec{v})$

Lenz!

← predznak?

U_i se upira vzroku svojega nastanka

Generator izm. nap.



$$\Phi_m = \vec{B} \cdot \vec{S} = BS \cos \varphi, \quad \varphi = \omega t$$

$$U_i = \underbrace{BS\omega}_{U_0} \sin \omega t$$

za tuljavo!

$$U_i = NBS\omega \sin \omega t$$

$$U_i = - \oint \vec{E}_i \cdot d\vec{s} \neq 0, \text{ če se } \vec{B} \text{ spreminja}$$

$$U_i = - \oint \vec{E}_i \cdot d\vec{s} = \frac{d\Phi_m}{dt} = \int \frac{d\vec{B}}{dt} \cdot d\vec{s}$$

$$\boxed{\oint \vec{E} \cdot d\vec{s} = - \int \frac{d\vec{B}}{dt} \cdot d\vec{s}}$$

Smer $d\vec{s}$ in $d\vec{S}$ se določi po pravilu desne roke

Posledica: $\vec{E}$ ni konzervativen
 $\rightarrow V(\vec{r})$ nima smisla

$$\left. \begin{array}{l} \text{Faraday} \\ \vec{B} \rightarrow \vec{E} \\ I(\vec{E}) \rightarrow \vec{B} \end{array} \right\} \in \mathcal{M}$$

Tuljava



Kondenzator

$$\underbrace{U \ell \ell \ell \ell \ell}_{L} \vec{s}$$

$$\begin{array}{c} e \\ | \\ | \\ | \\ | \\ | \\ d \end{array} \quad \begin{array}{c} s \end{array}$$

$$\text{Ampère: } \vec{B} = \frac{\mu_0 I}{l}$$

$$\text{Gauss: } \vec{E} = \frac{\sigma}{\epsilon_0} = \frac{e}{\epsilon_0 s}$$

$$\Phi_m = \vec{B} \cdot \vec{s} = LI$$

$$\Phi_e = e = CU$$

↑
kapaciteta

INDUKTIVNOST

$$L = \frac{N^2 \mu_0 s}{l}$$

$$C = \frac{\epsilon_0 s}{d}$$

$$U_c = - \frac{e}{C}$$

$$\text{Lastna indukcija: } U_i = - \frac{d\Phi_m}{dt} = - L \frac{dI}{dt}$$

$$(e = CU) \quad = \underbrace{\Phi_e d\Phi_e}$$

Polnjenje: I

$$\text{Polnjenje: } U \quad dA_e = U de = CU dU = \frac{1}{C} e de$$

(če hočemo napolniti tuljavo z min. izgubami) (če hočemo napolniti kond., min. izgubami)
 tabimo izvir s spremenljivim tokom) kapetastijo) moramo imeti izvir s spremenljivo

$$dA_m = U_I I dt = -L \frac{dI}{dt} I dt = -L I dI = -\frac{1}{2} \Phi_m d\Phi_m$$

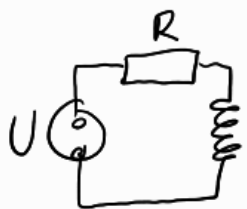
$$\boxed{W_m = A_m = \frac{1}{2} L I^2 = \frac{1}{2L} \Phi_m^2}$$

$$\boxed{W_e = \frac{1}{2} C U^2 = \frac{1}{2C} \Phi_e^2}$$



$$w_m = \frac{W_m}{V} = \frac{L I^2}{25l} = \frac{1}{2} \frac{B^2}{\mu_0}$$

Polnjenje/praznjenje L

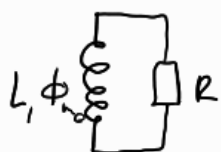
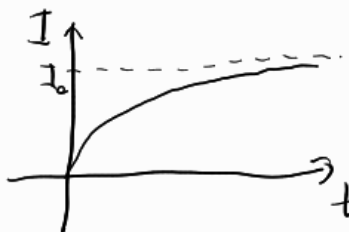


$$U_0 + U_L + U_R = 0$$

$$U_0 - L \frac{dI}{dt} - RI = 0$$

$$\int_0^t \frac{R}{L} dt = \int_0^t \frac{dI}{\frac{U_0}{R} - I} = -\ln \frac{I_0 - I}{I_0} = \frac{t}{\tau} \quad \boxed{\tau = \frac{L}{R}}$$

$$\boxed{I = I_0(1 - e^{-\frac{t}{\tau}})}$$

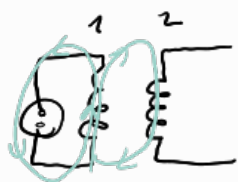


$$-L \frac{dI}{dt} = RI$$

$$\frac{dI}{I} = -\frac{dt}{\tau}$$

$$\rightarrow \boxed{I = I_0 e^{-\frac{t}{\tau}} \Rightarrow \Phi_m = \Phi_{m0} e^{-\frac{t}{\tau}}}$$

Vzajemna indukcija



$$U_{12} = -\frac{d\Phi_2}{dt}$$

$$\Phi_{m2} \propto I_1 \rightarrow \Phi_{m2} = L_{21} I_1$$

$$U_{11} = -L \frac{dI_1}{dt}$$

$$L = \begin{bmatrix} \text{lastne ind.} & \text{vzajemne} \\ \text{vzajemne} & \end{bmatrix}$$



$$\begin{aligned} \Phi_2 &= \Phi_1 \frac{S_2 N_2}{S_1 N_1} \\ &= \frac{N_1 \mu_0 S_1}{L_1} I_1 \frac{S_2 N_2}{S_1 N_1} \end{aligned}$$

$$\boxed{L_{21} = \frac{N_2 \mu_0 S_2 N_1}{L_1}}$$

$$L_{12} = L_{21}$$

teško dokazati,
ampak drži

Magn. polje v sholi

H atom



$$\vec{p}_n = I \vec{S}$$

$$\vec{B} (e, \vec{v} \rightarrow \vec{B})$$

$$(W_m = \vec{p}_n \cdot \vec{B})$$

spin

$$\vec{p}_n \rightarrow \vec{\mu} = \vec{\mu}_B = 10^{-23} \frac{J}{T}$$

$$N \mu_B \approx 10^4 A m^2$$

ker so atomi drugače obrnjeni - $\vec{p}_n \sim 0$

e kroži okrog jedra

$$\vec{p}_n = I \vec{S} = \frac{e_0}{t_0} \pi r^2 = \frac{e_0}{2} v r^2 \frac{m_e}{m_e} = \frac{e_0 \Gamma}{2 m_e} \sim \mu_B$$

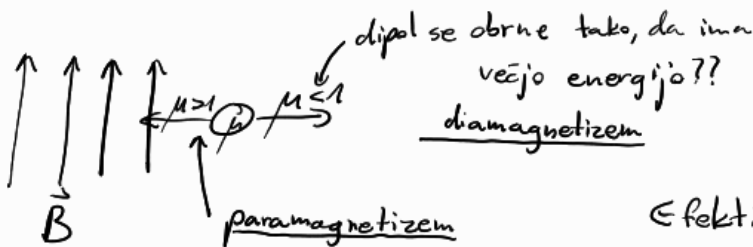
$$\mu_0 \rightarrow \mu \mu_0$$

permeabilnost

$$\text{gostota } B = \frac{\mu \mu_0 I}{2 \pi r}$$

$$\oint \vec{H} d\vec{s} = I$$

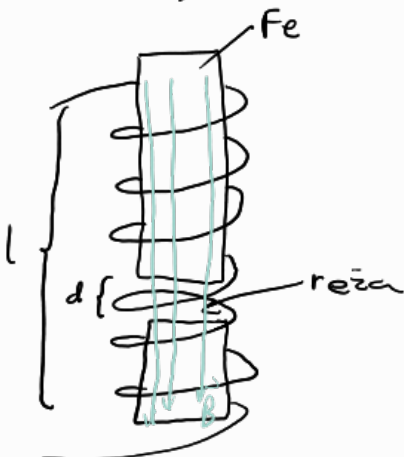
$$\text{jakost } H: \mu \mu_0 H = B$$



Efekti so majhni -

$$\text{urejenost: } e^{-\frac{W_m}{k_B T}} = \frac{\mu_B B}{k_B T} = \frac{10^{-23} J}{1,38 \cdot 10^{-23} \frac{J}{K} \cdot 300 K} \approx \frac{1}{400}$$

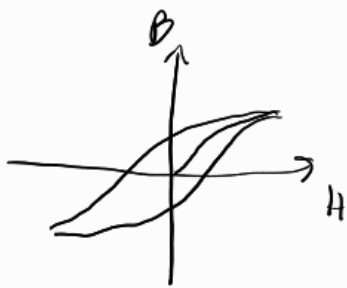
elektromagnet



$$\Phi_{m, \text{reža}} = \Phi_{m, Fe}$$

$$B_{\text{reža}} = B_{Fe}$$

$$\text{Ampere: } \oint \vec{H} d\vec{s} = NI = \frac{B}{\mu_0 \mu} (l-d) + \frac{B}{\mu_0} d \Rightarrow B \text{ je veliko manjši kot če ne bi bilo reže}$$



$$dA_e = U_{de} = U_i I dt = N S \frac{dB}{dt} \frac{Hl}{N} dt = S l H dB = V \cdot dw_m$$

$$U_i = \frac{d\Phi_m}{dt} = N S \frac{dB}{dt}$$

$$A = V w_m$$

merina: $U_i, I \propto H$
 $B \propto U_i dt$

Izmenični tok



$$W_e \rightarrow W_m \rightarrow W_e \rightarrow \dots$$

$$I=0 \quad I=I_{\max} \quad I=0$$

$$e=\max \quad e=0 \quad e=\max$$

↓
 ampak na
 drugi strani
 kond.

$$I = \frac{de}{dt} \quad v = \frac{dx}{dt}$$

$$F = ma$$

$$-kx = m\ddot{x}$$

Nihanje

$$x = x_0 \sin(\omega t + \phi)$$

$$\omega^2 = \frac{k}{m}$$

$$U_c + U_L = 0$$

$$-\frac{e}{C} - LI = 0$$

$$-\frac{e}{C} = L\ddot{e}$$

$$e = e_0 \sin(\omega t + \phi)$$

$$\omega^2 = \frac{1}{LC}$$

Dušenje



$$U_c + U_L + U_R = 0$$

$$+L \frac{dI}{dt} + IR + \frac{e}{C} = 0$$

$$L\ddot{e} + \dot{e}R + \frac{e}{C} = 0$$

$$\frac{d^2 e}{dt^2} + \frac{R}{L} \frac{de}{dt} + \frac{e}{LC} = 0$$

$$e = e_0 e^{-\beta t} \cos(\omega' t + \phi)$$

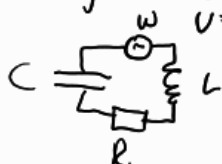
$$\beta = \frac{R}{2L}, \quad \omega_0^2 = \frac{1}{LC}, \quad \omega'^2 = \omega_0^2 - \beta^2$$

$$\frac{d^2 x}{dt^2} + 2\beta \frac{dx}{dt} + \omega_0^2 x = 0$$

$$x = x_0 e^{-\beta t} \cos(\omega' t + \phi)$$

$$\omega'^2 = \omega_0^2 - \beta^2$$

Vsiljeno nihanje



$$U = U_0 \sin(\omega t)$$

$$I = I_0 \sin(\omega t + \phi)$$

$$-\frac{\pi}{2} < \phi < \frac{\pi}{2}$$

Paskusi:

Vsak element posebej:

$$U = 8,5V$$

$$U_R = 8,5V$$

$$U_C = 7,5V$$

$$U_L = 5,5V$$

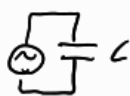
?

$$\text{[Circuit diagram: AC source, Resistor R]} \quad U + U_R = 0$$

$$U = RI$$

$$I = \frac{U}{R} = \frac{U_0}{R} \sin \omega t$$

$$\begin{aligned} \phi &= 0 \\ I_0 &= \frac{U_0}{R} \end{aligned}$$



$$U + U_C = 0$$

$$U = \frac{q}{C} = U_0 \sin \omega t$$

$$I = \frac{dq}{dt} = C U_0 \omega \cos \omega t$$

$$\begin{aligned} \phi &= \frac{\pi}{2} \\ I_0 &= C U_0 \omega \\ R_C &= \frac{1}{\omega C} \end{aligned}$$

tak prehiteva napetost



$$U + U_L = 0$$

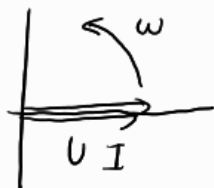
$$U = L \frac{dI}{dt}$$

$$I = \frac{1}{L} \int U dt = -\frac{U_0}{L\omega} \cos \omega t$$

$$\begin{aligned} \phi &= -\frac{\pi}{2} \\ I_0 &= \frac{U_0}{L\omega} \\ R_L &= \omega L \end{aligned}$$

tak zaostaja za nap.

R:



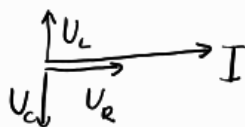
kazalni diagram

(če gledamo projekcijo U in I na katerikoli os dobimo nihanje)

C:



Vsi skupaj:



→ v paskusu:

$$U_R^2 + (U_L - U_C)^2 = 7V$$

tuljava ima še neko upornost

L:



Če so vezja kompleksna - težko

⇒ $R_L, R_C, R_R \rightarrow Z_L, Z_C, Z_R$
upor impedanca
(kompleksno št.)

$$U = ZI$$

$$Z_L = i\omega L$$

$$Z_C = \frac{1}{i\omega C} = -\frac{i}{\omega C}$$

$$U = U_0 \cos \omega t \rightarrow U = U_0 e^{i\omega t}$$

$$I = I_0 \cos(\omega t + \delta) \rightarrow I = I_0 e^{i(\omega t + \delta)} = I_0 e^{i\omega t} e^{i\delta}$$



$$U = ZI$$

$$U_0 e^{i\omega t} = Z I_0 e^{i\omega t} e^{i\delta}$$

$$Z = \frac{U_0}{I_0} e^{i\delta}$$

od tu naprej rešujemo z impedanco

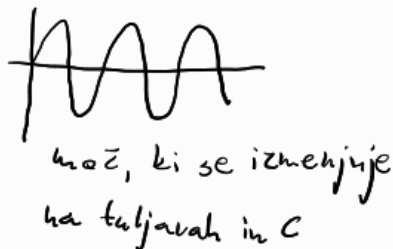
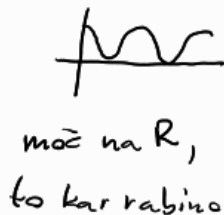
kot bi reševali z upori

(zaporedno: $Z = Z_1 + Z_2$)

(vporedno: $\frac{1}{Z} = \frac{1}{Z_1} + \frac{1}{Z_2}$)

$$P = UI = U_0 I_0 \cos(\omega t) \cos(\omega t + \delta) (\cos \omega t \cos \delta - \sin \omega t \sin \delta) =$$

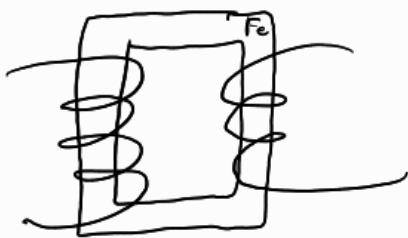
$$= U_0 I_0 (\underbrace{\cos^2 \omega t \cos \delta} - \underbrace{\cos \omega t \sin \omega t \sin \delta})$$



$$\bar{P} = \frac{1}{2} U_0 I_0 \cos \delta = \underline{U_{ef} I_{ef} \cos \delta}$$

$$\underline{U_{ef} = \frac{U_0}{\sqrt{2}}, I_{ef} = \frac{I_0}{\sqrt{2}}}$$

Transformator



vse silnice (večina)
skozi obe tuljavi

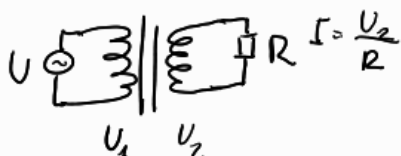
$$\Phi_{m1} = N_1 \int \vec{B} \cdot d\vec{S}$$

$$\Phi_{m2} = N_2 \int \vec{B} \cdot d\vec{S}$$

$$U_1 = \dot{\Phi}_{m1} = N_1 \int \dot{\vec{B}} \cdot d\vec{S}$$

$$U_2 = \dot{\Phi}_{m2} = N_2 \int \dot{\vec{B}} \cdot d\vec{S}$$

$$\rightarrow \boxed{\frac{U_1}{U_2} = \frac{N_1}{N_2}}$$



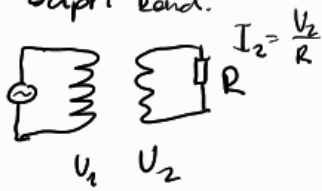
$$P_1 = P_2$$

$$U_1 I_1 = U_2 I_2$$

$$\boxed{\frac{I_1}{I_2} = \frac{N_2}{N_1}}$$

Idealen transformator

odprt kond.



El. in mag. polje

Pretok:

- Gauss: $\oint \vec{D} \cdot d\vec{s} = e$

$$\oint \vec{B} \cdot d\vec{s} = 0$$

Indukcija:

- Faraday: $\oint \vec{E} \cdot d\vec{s} = - \int \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s}$

- Ampere: $\oint \vec{H} \cdot d\vec{s} = I$

Maxwellove enačbe:

